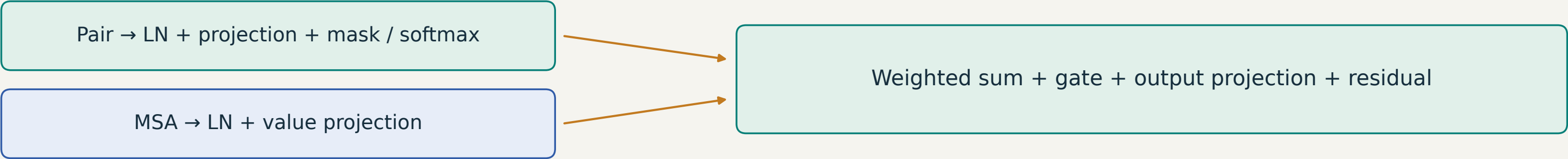


Pair-Weighted Averaging: join two reusable input branches

Independent pair-weight and MSA-value paths feed a fused output computation [C6]

A / EXECUTION AND FUSION



B / ALGORITHM AND DATA REUSE

PAIR WEIGHTS

$$P_{ijh} = \text{softmax}_j(b_{ijh})$$

$$U_{sihd} = \sum_j P_{ijh} V_{sjhd}$$

Pair weights are reused across MSA rows **s**.

Gate and output projection consume the weighted result in the fused path.

MSA VALUES

$$V_{sjhd}$$

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384

4.161 → 1.561 ms

2.67x

Baseline: Own engine path (job 15737)

Shape: S1024; D_msa64 / D_pair128

Arrows show independent inputs. Training mask preparation and multi-kernel backward are outside the forward boxes.